

MPONLINE
Advanced Software Engineering &
Development Internship
Batch 11(A)

PROJECT REPORT

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CHAPTER – 1

INTRODUCTION

Libraries play an important role in educational institutions by providing access to books, journals, magazines, research papers, and other learning resources. Traditionally, library operations such as maintaining book records, issuing books, returning books, and managing student information were performed manually. Manual systems consume considerable time, require extensive paperwork, and increase the chances of human error.

The Library Management System is designed to digitize these operations and provide an efficient mechanism for managing library resources. The application enables administrators to maintain complete information about books, students, librarians, and borrowing records from a centralized platform.

The project has been developed using modern web technologies such as ASP.NET Core MVC, Entity Framework Core, and SQL Server. These technologies provide a secure, scalable, and high-performance environment for application development.

The system follows the Model–View–Controller (MVC) architecture, where the Model manages application data, the View provides the graphical user interface, and the Controller processes user requests and coordinates communication between the Model and View.

The application includes several functional modules such as login management, dashboard, book management, student management, librarian management, borrowing management, return management, and contact support.

By replacing manual record maintenance with computerized operations, the system significantly improves efficiency, accuracy, and security while reducing administrative workload.

CHAPTER – 2

OBJECTIVES OF THE PROJECT

The primary objective of the Library Management System is to automate and simplify the management of library activities by replacing traditional manual processes with a computerized solution.

The specific objectives of the system are:

1. To Maintain Book Records Efficiently

The system should allow administrators to add, edit, delete, and view books stored in the library. Each book record contains information such as title, author, ISBN number, publication date, and availability status.

2. To Automate Borrowing and Returning Operations

The system should provide a mechanism to issue books to students and record return transactions automatically while maintaining complete borrowing history.

3. To Reduce Manual Errors

By storing data electronically, the chances of duplicate entries, lost records, and calculation mistakes are significantly minimized.

4. To Improve Data Security

The login module ensures that only authorized users can access the system and perform administrative operations.

5. To Maintain Student and Librarian Information

The system provides separate modules for managing student details and librarian information in an organized manner.

6. To Increase Operational Efficiency

The computerized system reduces paperwork and enables faster searching, updating, and retrieval of records.

7. To Provide Real-Time Availability Information

The availability status of books can be tracked instantly, ensuring that administrators know whether a book is available or already borrowed.

CHAPTER – 3

TECHNOLOGY STACK

3.1 Introduction

The Library Management System has been developed using modern web technologies that provide a secure, scalable, and user-friendly environment. The project follows the Model–View–Controller (MVC) architecture, ensuring proper separation of application logic, user interface, and data management. The selected technologies enable efficient development, easy maintenance, and better performance.

3.2 Front-End Technologies

HTML5

HTML5 is used to design the structure of web pages. It provides semantic elements for creating forms, tables, navigation menus, and other user interface components.

CSS3

CSS3 is used to enhance the visual appearance of the application. It provides styling for layouts, colors, fonts, spacing, and responsive design.

Bootstrap 5

Bootstrap is a front-end framework used to create responsive and mobile-friendly web pages. It provides pre-designed components such as buttons, forms, navigation bars, tables, cards, and alerts, making the application visually appealing and easy to use.

3.3 Back-End Technologies

ASP.NET Core MVC (.NET 8)

ASP.NET Core MVC is the primary framework used for developing the application. It follows the Model–View–Controller architecture, which separates business logic, user interface, and data access into independent components. This architecture improves maintainability, scalability, and code organization.

C#

C# is the programming language used to implement the application's business logic. It is an object-oriented language that provides features such as exception handling, strong type checking, and robust security, making it suitable for enterprise-level applications.

3.4 Database

Microsoft SQL Server

Microsoft SQL Server is used as the backend database for storing application data. It maintains information related to books, students, librarians, user login credentials, and borrowing records. SQL Server ensures secure, reliable, and efficient data management.

3.5 Object Relational Mapping (ORM)

Entity Framework Core

Entity Framework Core is an Object Relational Mapper (ORM) that simplifies database operations by allowing developers to work with C# objects instead of writing complex SQL queries. It supports CRUD (Create, Read, Update, Delete) operations and follows the Code First development approach.

3.6 Development Environment

Component	Technology Used
Programming Language	C#
Framework	ASP.NET Core MVC (.NET 8)
Front-End	HTML5, CSS3, Bootstrap 5
Database	Microsoft SQL Server
ORM	Entity Framework Core
IDE	Visual Studio 2022
Version Control	Git (Optional)
Web Browser	Google Chrome / Microsoft Edge
Operating System	Windows 10 / Windows 11

3.7 MVC Architecture

The project follows the Model–View–Controller (MVC) architecture.

- Model: Manages application data and interacts with the SQL Server database through Entity Framework Core.
- View: Provides the graphical user interface using Razor Views, HTML5, CSS3, and Bootstrap.
- Controller: Handles user requests, processes business logic, and coordinates communication between the Model and the View.

3.8 Advantages of the Selected Technology Stack

- Fast and secure web application development.
- Clear separation of concerns using MVC architecture.
- Easy database integration with Entity Framework Core.
- Responsive and user-friendly interface using Bootstrap.
- Scalable architecture suitable for future enhancements.
- Reliable data storage using SQL Server.
- Easy maintenance and code reusability.
- Support for validation and error handling.

CHAPTER 4

REQUIREMENT ANALYSIS

Requirement analysis is the process of identifying the needs and expectations of users and converting them into system requirements. It helps developers understand what features should be implemented and how the system should behave.

For the Library Management System, requirement analysis was performed before the implementation phase to determine the necessary modules and functionalities.

4.1 Functional Requirements

Functional requirements describe the operations that the system must perform.

FR-1: User Login

The system shall allow authorized users to log in using valid credentials.

FR-2: Dashboard

The system shall display overall statistics regarding books, students, and librarians.

FR-3: Book Management

The administrator shall be able to:

- Add books.
- Edit book details.
- Delete books.
- View book information.
- Check availability status.

FR-4: Student Management

The administrator shall be able to:

- Add students.
- Edit student records.
- Delete students.

FR-5: Librarian Management

The administrator shall be able to:

- Add librarian details.
- Edit librarian information.
- Delete librarian records.
- View librarian information.

FR-6: Borrow Book

The administrator shall be able to issue books to students by creating borrowing records.

FR-7: Return Book

The administrator shall be able to return borrowed books and update their availability status.

FR-8: Data Validation

The system shall validate all mandatory fields before accepting input.

FR-9: Database Storage

The system shall permanently store all information inside SQL Server.

4.2 Non-Functional Requirements

- 1. Performance-** The system should retrieve records quickly and efficiently.
- 2. Security-** Only authenticated users should access administrative features.
- 3. Reliability** - The system should maintain accurate and consistent records.
- 4. Maintainability-** The system should allow future modifications with minimal effort.
- 5. Scalability** - The application should support future expansion and additional modules.
- 6. Usability** - The user interface should be simple and easy to understand.
- 7. Availability** -The application should remain accessible whenever required.

4.3 Hardware Requirements

Component	Minimum Requirement
Processor	Intel Core i3
RAM	4 GB
Storage	50 GB
Monitor	1366 × 768 Resolution
Keyboard	Standard Keyboard
Mouse	Optical Mouse

4.4 Software Requirements

Software	Version
Operating System	Windows 10/11
IDE	Visual Studio 2022
Framework	ASP.NET Core MVC (.NET 8)
Language	C#
Database	SQL Server
ORM	Entity Framework Core
Frontend	HTML5, CSS3, Bootstrap
Browser	Google Chrome / Microsoft Edge

4.5 Feasibility Analysis

1. Technical Feasibility

The technologies used in the project are modern, open-source, and well-supported. ASP.NET Core and SQL Server provide sufficient capabilities for implementing all required features.

2. Operational Feasibility

The application is user-friendly and requires minimal training for administrators.

3. Economic Feasibility

The software can be developed using freely available tools such as Visual Studio Community Edition and SQL Server Express, reducing overall development costs.

CHAPTER -5

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

5.1 Introduction

The Software Requirement Specification defines the complete behaviour and requirements of the Library Management System. It acts as a blueprint for the development process.

5.2 Modules

The purpose of the Library Management System is to automate library activities including:

- Authentication & Session Management
- Dashboard Module
- Book Module
- Newspaper Module
- Student Module
- Librarian Module
- Borrow & Return Module
- Availability tracking.
- Profile Module
- About Us & Contact Us

5.3 Product Perspective

The Library Management System is a standalone web application developed using ASP.NET Core MVC and SQL Server. It follows the MVC architecture and stores all information inside a centralized database.

5.4 Product Functions

The system performs the following functions:

- User authentication.
- Dashboard monitoring.
- Book management.

- Student management.
- Librarian management.
- Borrowing management
- Return management.
- Availability tracking.
- Database operations.
- Validation and exception handling.

5.5 User Characteristics

Administrator

Has complete access to the application and can manage all modules.

Librarian

Can manage books, borrowing records, and return records.

Student

Can borrow books through administrative approval.

5.6 Constraints

- Requires internet browser support.
- Requires SQL Server installation.
- Requires .NET runtime environment.
- Requires valid login credentials.
- Requires proper database connectivity.

CHAPTER -6 SYSTEM DESIGN

6.1 System Architecture

The Library Management System follows the Model–View–Controller (MVC) architecture.

The application is divided into three major components:

- Model – Handles business logic and database interaction.
- View – Provides the user interface.
- Controller – Processes user requests and communicates between the Model and View.

The database stores all records related to books, students, librarians, borrowing transactions, and login credentials.

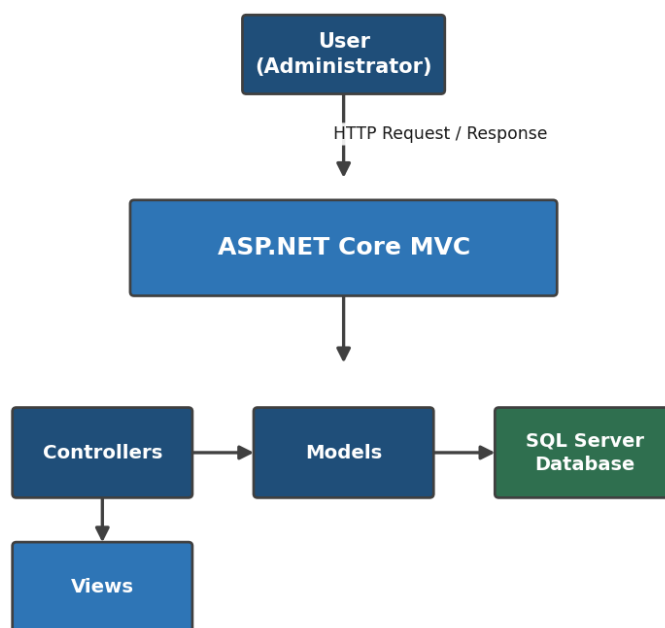


Fig. 6.1 System Architecture Diagram

6.2 ER Diagram

The database contains five major entities:

- Login
- Books
- BorrowRecords
- Students

- Librarians

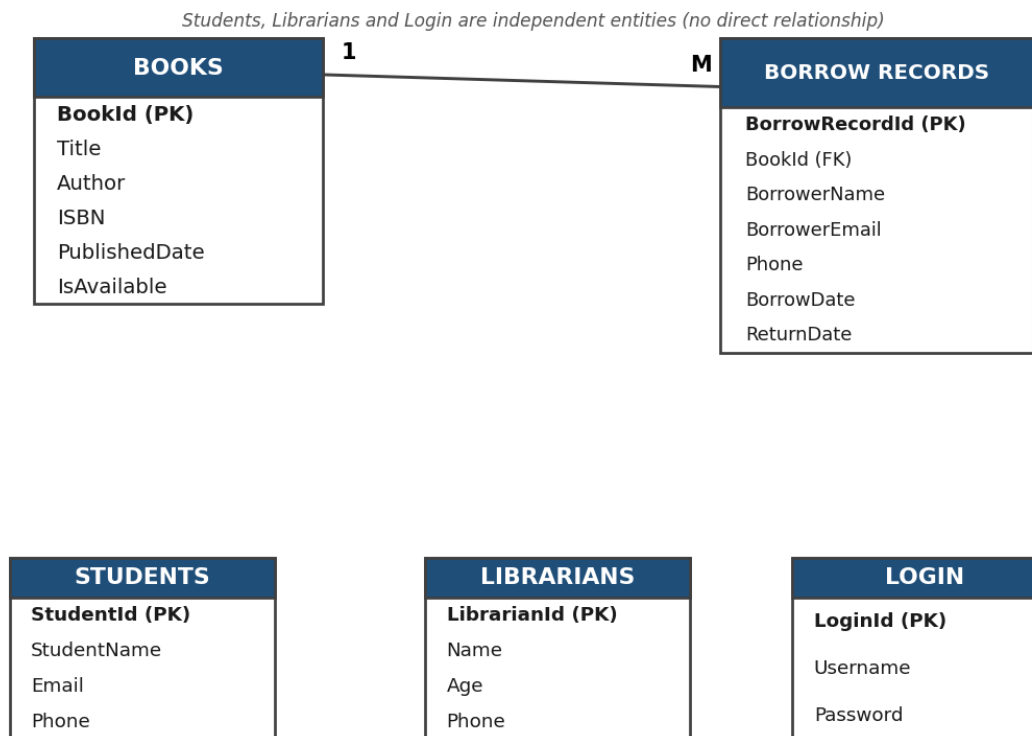


Fig. 6.2 Entity-Relationship (ER) Diagram

6.3 Data Flow Diagram (Level 0)

The Level 0 DFD represents the overall interaction between the administrator and the Library Management System.

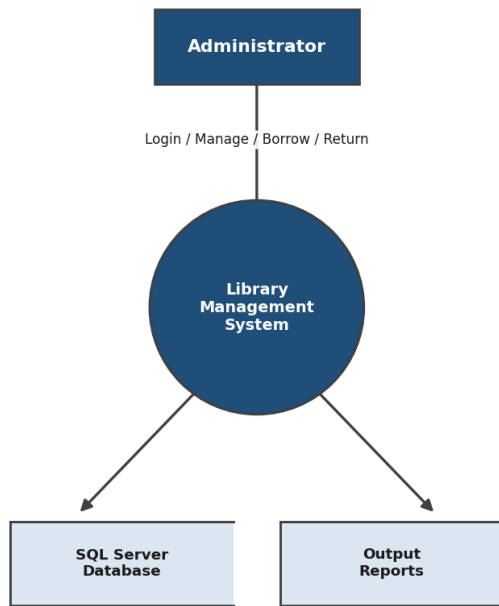


Fig. 6.3 Data Flow Diagram - Level 0

6.4 Data Flow Diagram (Level 1)

The Level 1 DFD expands the single process of Level 0 into the individual modules of the system — login authentication, dashboard, book management, student management, librarian management, and the borrow/return process — all of which read from and write to the central database.

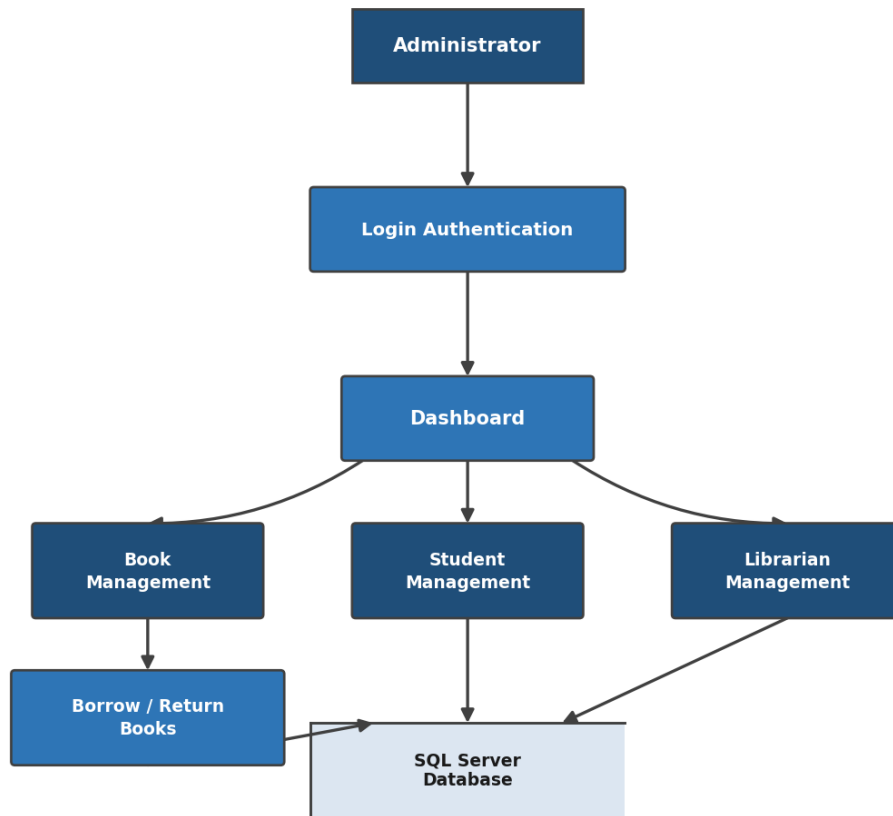


Fig. 6.4 Data Flow Diagram - Level 1

6.5 Use Case Diagram

The Administrator is the sole actor of the system and can perform the following use cases: Login, Manage Books, Manage Students, Manage Librarians, Borrow Books, Return Books, and View Dashboard.

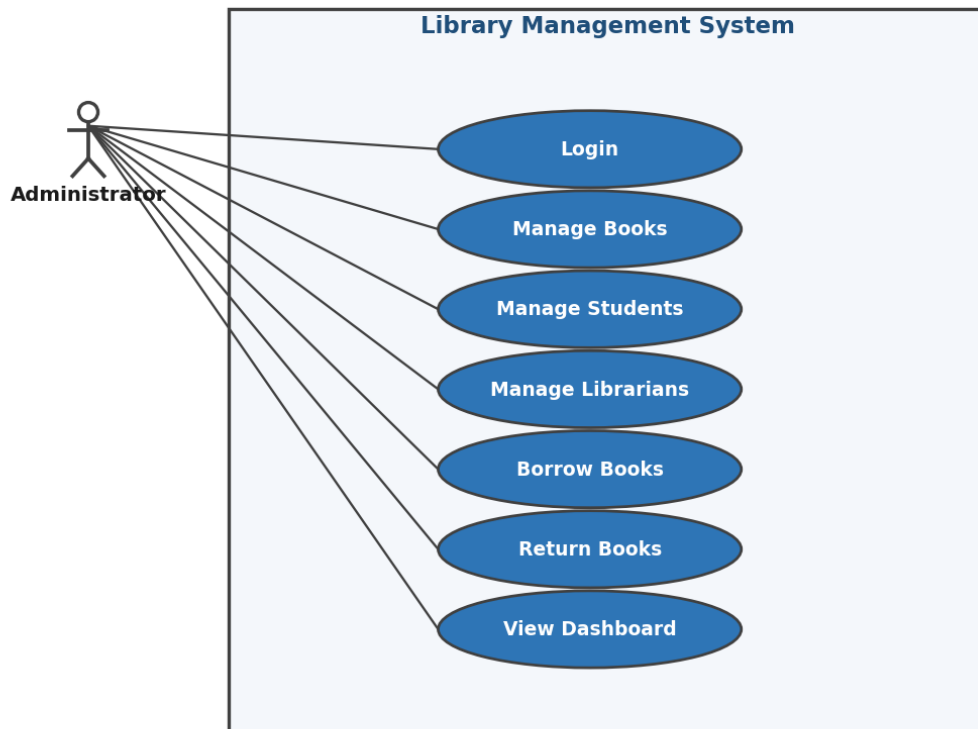


Fig. 6.5 Use Case Diagram

6.6 Class Diagram

The class diagram below models the core entities of the system and the association between the Book and BorrowRecord classes.

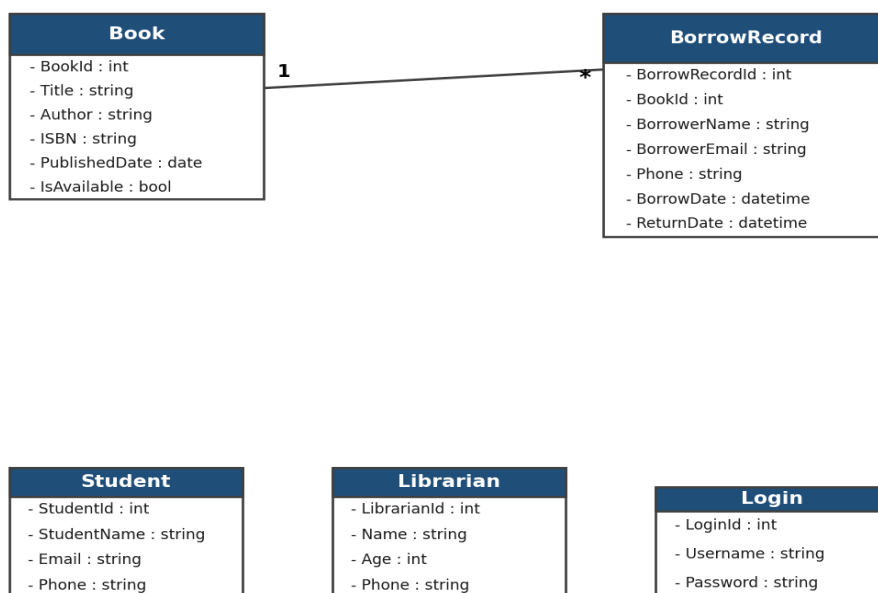


Fig. 6.6 Class Diagram

6.7 Sequence Diagram (Borrow Book)

The sequence diagram illustrates the interaction between the Administrator, the System, and the Database during the process of logging in and borrowing a book.

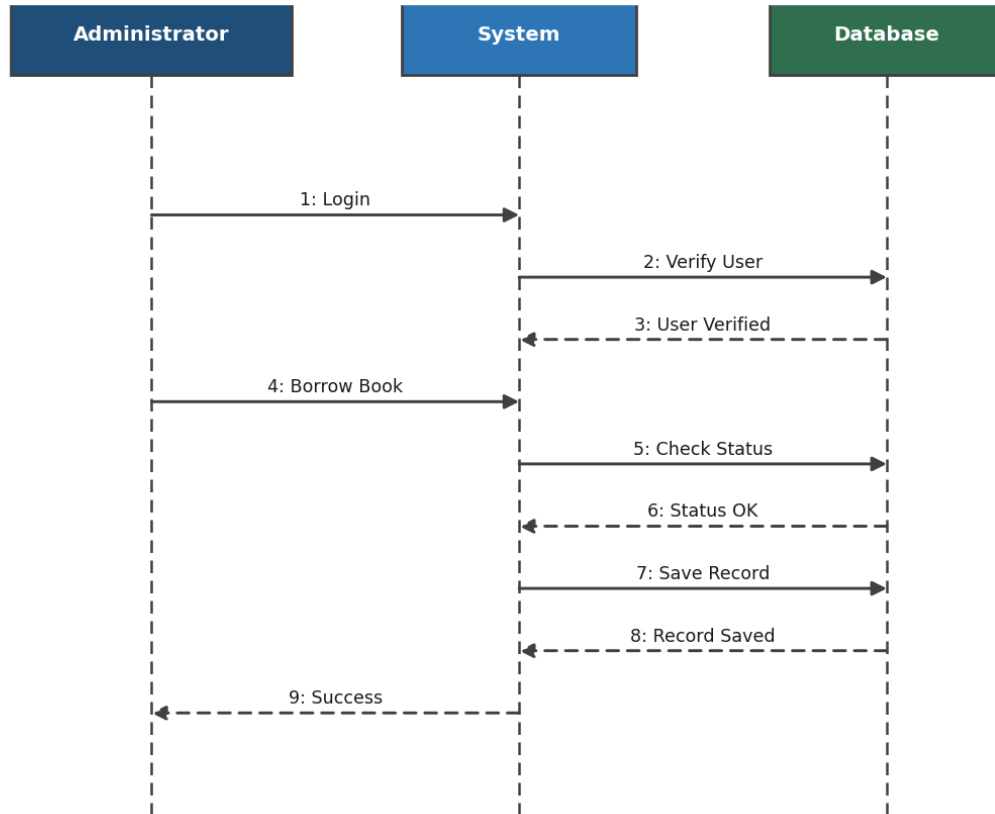


Fig. 6.7 Sequence Diagram - Borrow Book

CHAPTER -7

DATABASE DESIGN

7.1 Introduction

The database is one of the most important components of the Library Management System. It stores all information related to books, students, librarians, user login credentials, and borrowing transactions.

The project uses Microsoft SQL Server as the backend database and Entity Framework Core (Code First Approach) for database operations. Entity Framework automatically creates tables based on the model classes and maintains relationships among them.

The database has been designed to minimize redundancy, ensure data integrity, and support efficient retrieval of records.

7.2 Database Features

- Centralized storage of information
- Data consistency
- Fast searching and retrieval
- Secure storage of records
- Relationship management using primary and foreign keys
- Easy backup and recovery
- Supports CRUD operations

7.3 Database Tables

The Library Management System consists of the following tables:

Table Name	Purpose
Login	Stores login credentials
Books	Stores book information
BorrowRecords	Stores issue and return transactions
Students	Stores student details
Librarians	Stores librarian details

7.4 Books Table

Table Name: Books

Field Name	Data Type	Size	Constraint
BookId	int	-	Primary Key
Title	nvarchar	100	Not Null
Author	nvarchar	100	Not Null
ISBN	nvarchar	20	Unique
PublishedDate	date	-	Not Null
IsAvailable	bit	-	Default True

Description: The Books table stores complete information about every book available in the library. Each book has a unique BookId and ISBN number.

7.5 BorrowRecords Table

Table Name: BorrowRecords

Field Name	Data Type	Constraint
BorrowRecordId	int	Primary Key
BookId	int	Foreign Key
BorrowerName	nvarchar(100)	Not Null
BorrowerEmail	nvarchar(100)	Not Null
Phone	nvarchar(15)	Not Null
BorrowDate	datetime	Default Current Date
ReturnDate	datetime	Nullable

Description: This table records every borrowing transaction. It maintains complete information about issued books and their return status.

7.6 Students Table

Field Name	Data Type	Constraint
StudentId	int	Primary Key
StudentName	nvarchar(100)	Not Null
Email	nvarchar(100)	Not Null
Phone	nvarchar(15)	Not Null

Description: This table stores information about registered students who borrow books.

7.7 Librarians Table

Field Name	Data Type	Constraint
LibrarianId	int	Primary Key
Name	nvarchar(100)	Not Null
Age	int	Not Null
Phone	nvarchar(15)	Not Null

Description: This table stores librarian information maintained by the administrator.

7.8 Login Table

Field Name	Data Type	Constraint
LoginId	int	Primary Key
Username	nvarchar(50)	Unique
Password	nvarchar(50)	Not Null

Description: The Login table stores authentication credentials used to access the Library Management System.

7.9 Relationship Between Tables

The primary relationship in the system is between the Books table and the BorrowRecords table.

- One book can have multiple borrow records over time.
- Each borrow record belongs to only one book.
- BookId acts as the Foreign Key in BorrowRecords.

Books (1) ———< BorrowRecords (Many)

Students, Librarians, and Login are independent tables with no direct foreign-key relationship to the other entities.

CHAPTER -8

MODULE DESCRIPTION

The Library Management System has been divided into several independent modules. Each module performs a specific task and collectively contributes to the smooth functioning of the application.

8.1 Login Module

Purpose: The Login Module authenticates users before granting access to the system.

Functions:

- User Login
- Credential Verification
- Redirect to Dashboard
- Invalid Login Handling

Input:

- Username
- Password

Output:

- Dashboard
- Login Failed Message

8.2 Dashboard Module

Purpose: The Dashboard provides a quick overview of library activities.

Functions:

- Display total books
- Display total students
- Display total librarians
- Navigation to all modules

Output:

- Statistical information
- Quick navigation links

8.3 Book Management Module

Purpose: This module manages all book-related activities.

Functions:

- Add Book
- Edit Book
- Delete Book
- View Book Details
- Search Books
- Check Availability

Input:

- Title
- Author
- ISBN
- Publication Date

Output:

- Updated Book List

8.4 Student Management Module

Purpose: Maintains student information.

Functions:

- Add Student
- Edit Student
- Delete Student
- View Students

Input:

- Student Name
- Email
- Phone Number

Output:

- Updated Student Records

8.5 Librarian Module

Purpose: Stores librarian details.

Functions:

- Add Librarian
- Edit Librarian
- Delete Librarian
- View Librarians

8.6 Borrow Book Module

Purpose: Allows administrators to issue books.

Functions:

- Select Book
- Enter Borrower Details
- Create Borrow Record
- Update Availability

Input:

- Borrower Name
- Email
- Phone Number

Output:

- Book Successfully Borrowed

8.7 Return Book Module

Purpose: Handles book returns.

Functions:

- Return Book
- Update Return Date
- Mark Book Available

Output:

- Book Successfully Returned

8.8 Contact Module

Purpose: Provides contact information and communication support.

Functions:

- Display Contact Details
- Help Information
- User Assistance

CHAPTER – 9

IMPLEMENTATION

9.1 Development Environment

Component	Technology
IDE	Visual Studio 2022
Framework	ASP.NET Core MVC
Programming Language	C#
Database	SQL Server
ORM	Entity Framework Core
Frontend	HTML5, CSS3, Bootstrap
Version Control	Git (optional)

9.2 Project Structure

```
LibraryManagementSystem
├── Controllers
│   ├── LoginController
│   ├── BooksController
│   ├── StudentController
│   ├── LibrarianController
│   ├── BorrowController
│   └── DashboardController
├── Models
│   ├── Book
│   ├── BorrowRecord
│   ├── Student
│   ├── Librarian
│   ├── Login
│   └── LibraryContext
├── Views
│   ├── Login
│   ├── Books
│   ├── Students
│   ├── Librarians
│   ├── Borrow
│   └── Dashboard
├── ViewModels
│   └── BorrowViewModel
```

```
| ReturnViewMode
|— wwwroot
|   CSS
|   JavaScript
|   Images
|— appsettings.json
```

CHAPTER – 10

TESTING

10.1 Introduction

Testing is the process of verifying that the software performs according to the specified requirements. It helps identify errors, improves reliability, and ensures that the application functions correctly before deployment.

The Library Management System was tested using different test cases to validate each module.

10.2 Types of Testing

Unit Testing

Each controller and model was tested independently to verify correct functionality.

Integration Testing

The interaction between controllers, views, models, and the SQL Server database was tested.

System Testing

The complete application was tested to ensure all modules worked together correctly.

Validation Testing

Input validation was performed to ensure users enter valid information.

User Acceptance Testing

The application was tested from an administrator's perspective to confirm that it met project requirements.

10.3 Test Cases

TC ID	Module	Input / Objective	Expected Result	Status
TC-01	Login	Valid Username & Password	Dashboard opens	Pass
TC-02	Books	Valid Book Details	Book Added Successfully	Pass
TC-03	Books	Edit existing book details	Book Updated	Pass
TC-04	Books	Delete selected book	Book Deleted	Pass
TC-05	Students	Valid Student Details	Student Added	Pass

TC ID	Module	Input / Objective	Expected Result	Status
TC-06	Borrow	Valid Borrower Details	Borrow Record Created	Pass
TC-07	Return	Return an issued book	Book Returned	Pass
TC-08	Forms	Invalid / empty form input	Validation Message Displayed	Pass

CHAPTER -11

OUTPUT SCREENS

The following figures present the application screenshots along with a short description of each screen.

Figure 11.1 – Login Page

The Login page authenticates administrators before allowing access to the Library Management System.

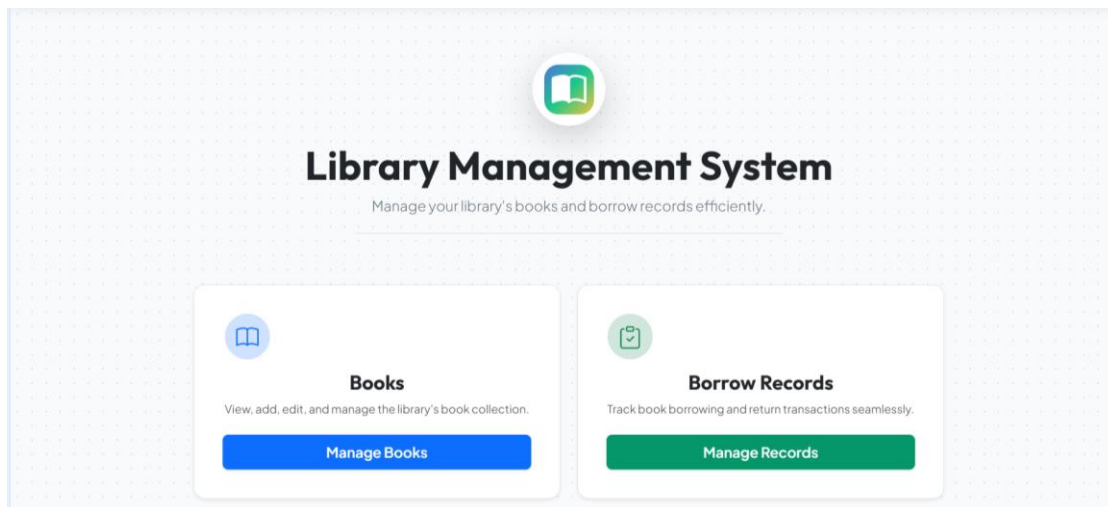


Figure 11.2 – Dashboard

The Dashboard provides an overview of books, students, librarians, and navigation to different modules.

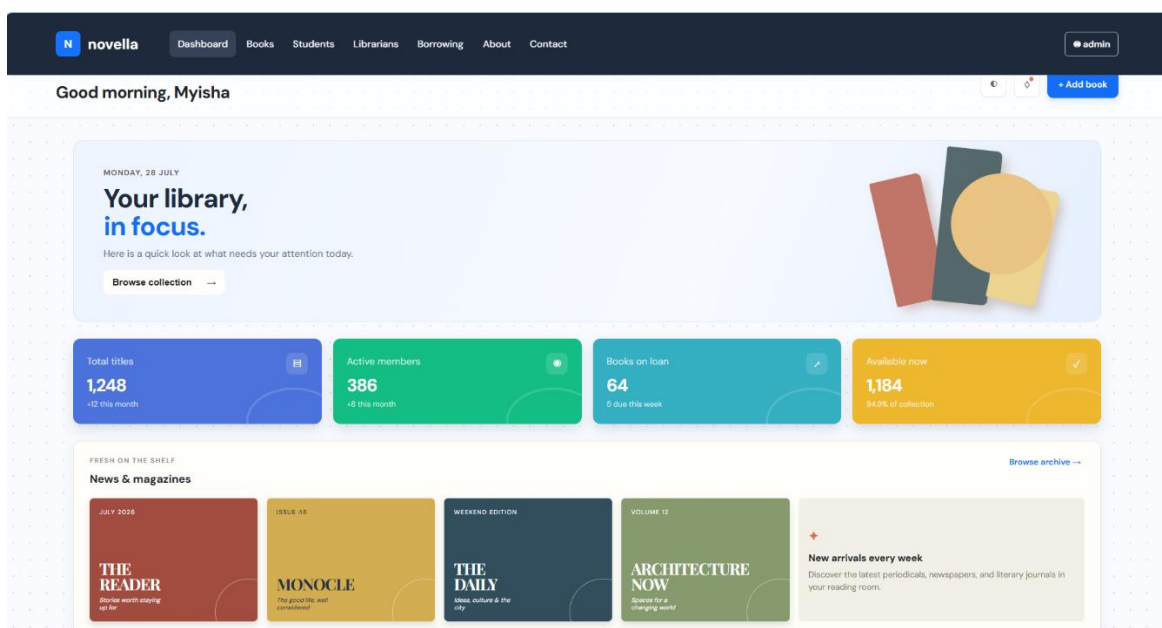


Figure 11.3 – Books Module

Displays all books with options to add, edit, delete, view details, borrow, and return books.

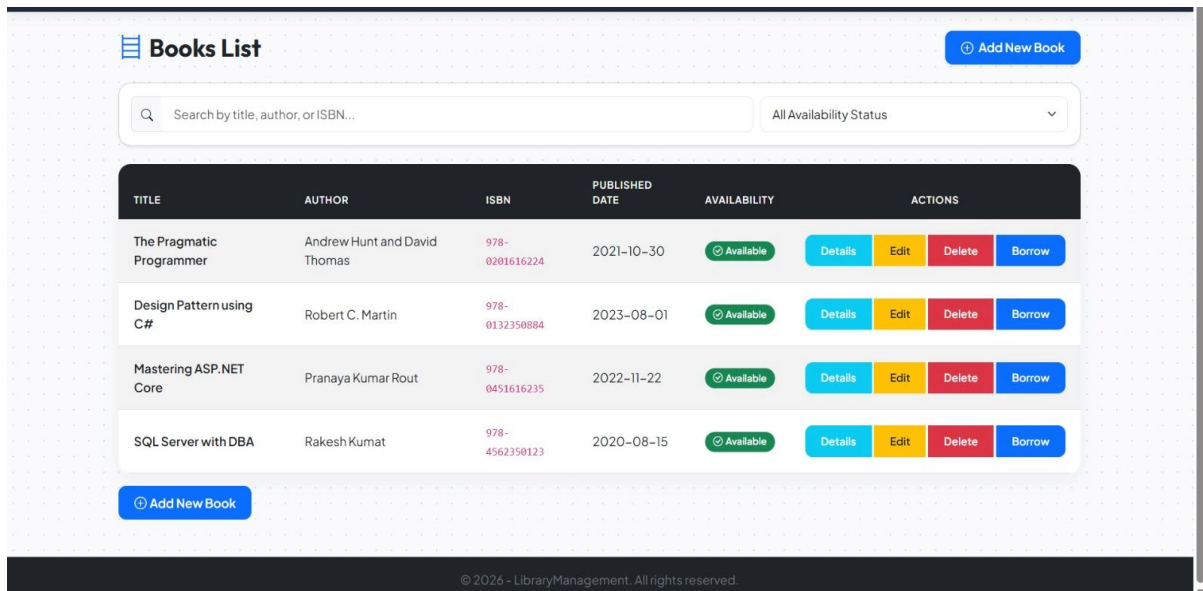


Figure 11.4 – Add Book

Allows the administrator to add a new book by entering title, author, ISBN, and publication date.

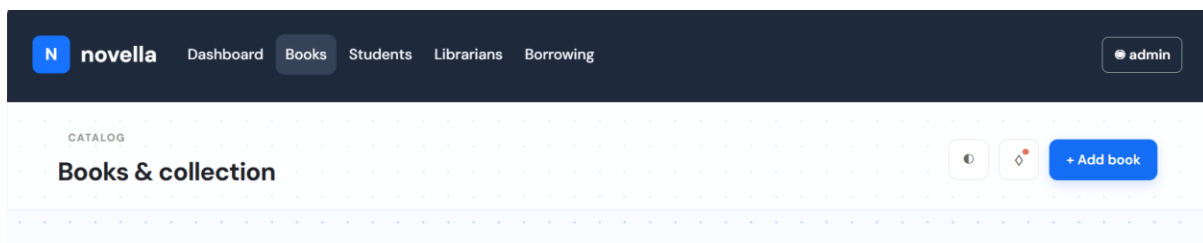


Figure 11.5 – Book Details

Shows complete information about a selected book.

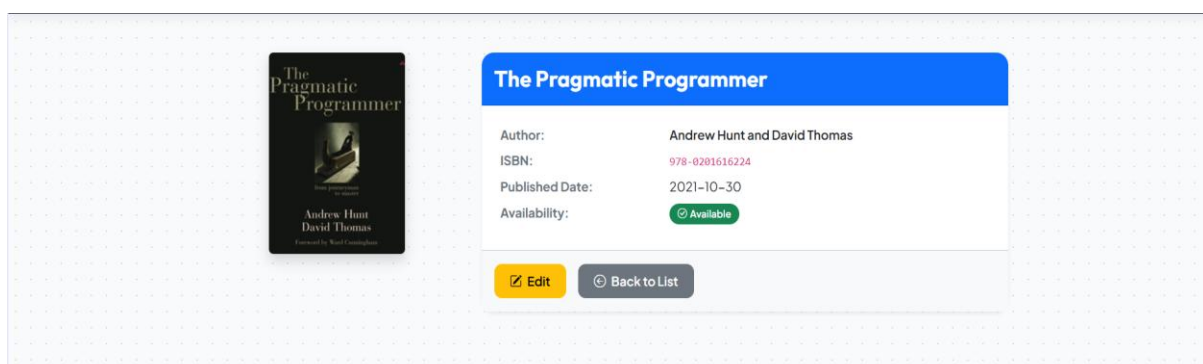


Figure 11.6– Student Module

Displays the list of registered students and supports CRUD operations.

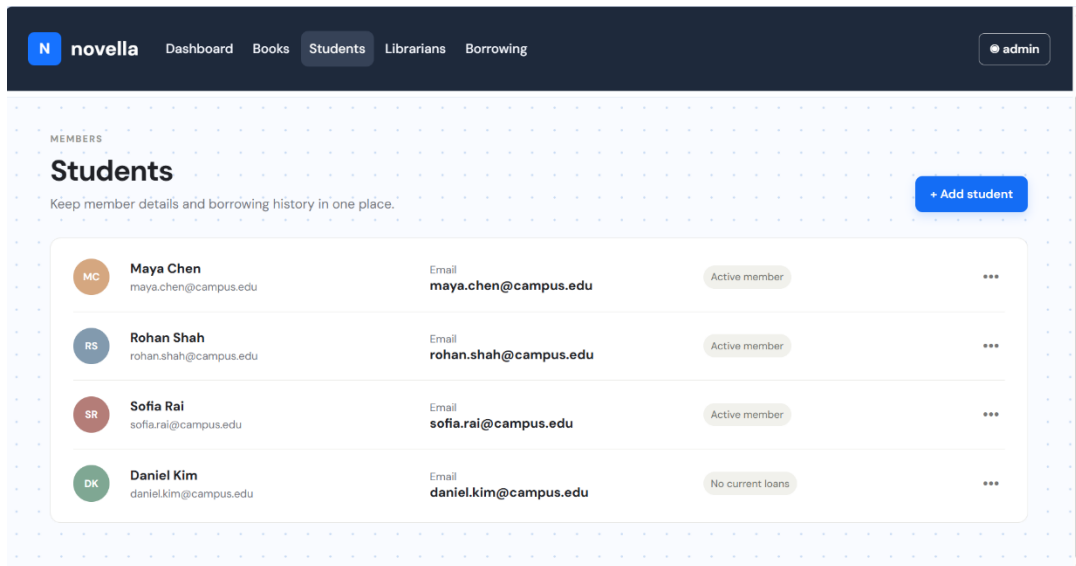


Figure 11.7 – Librarian Module

Displays librarian information and allows management of librarian records.

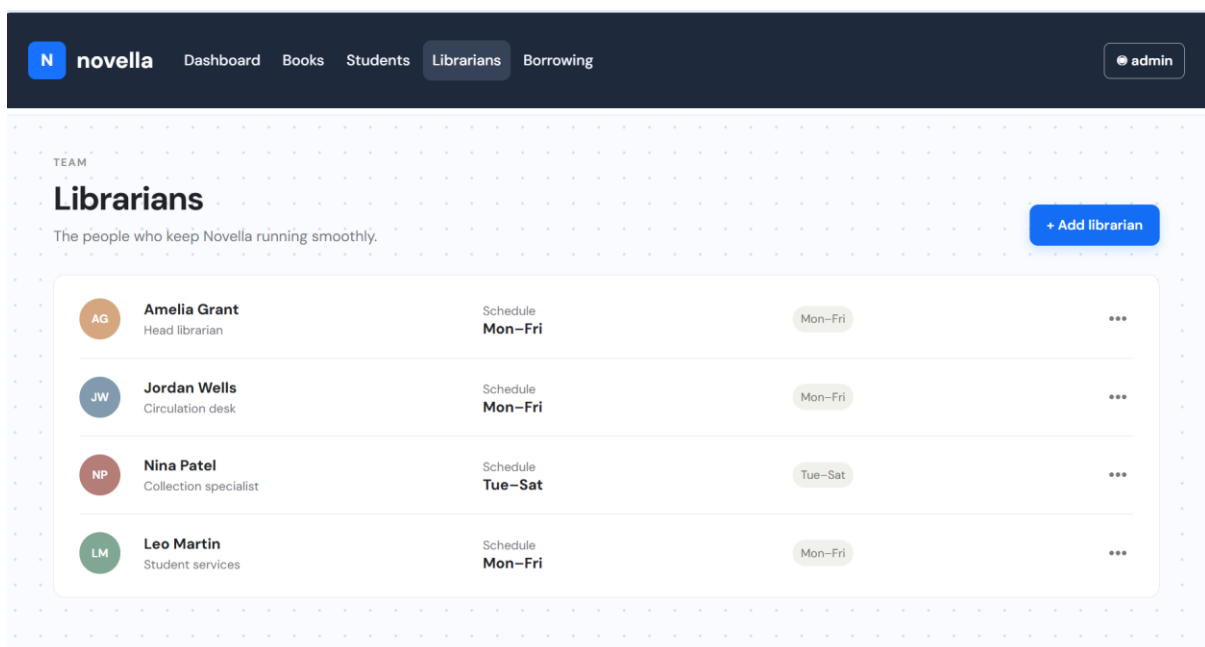


Figure 11.8 – Borrow Book

Allows the administrator to issue books to borrowers by entering borrower details.

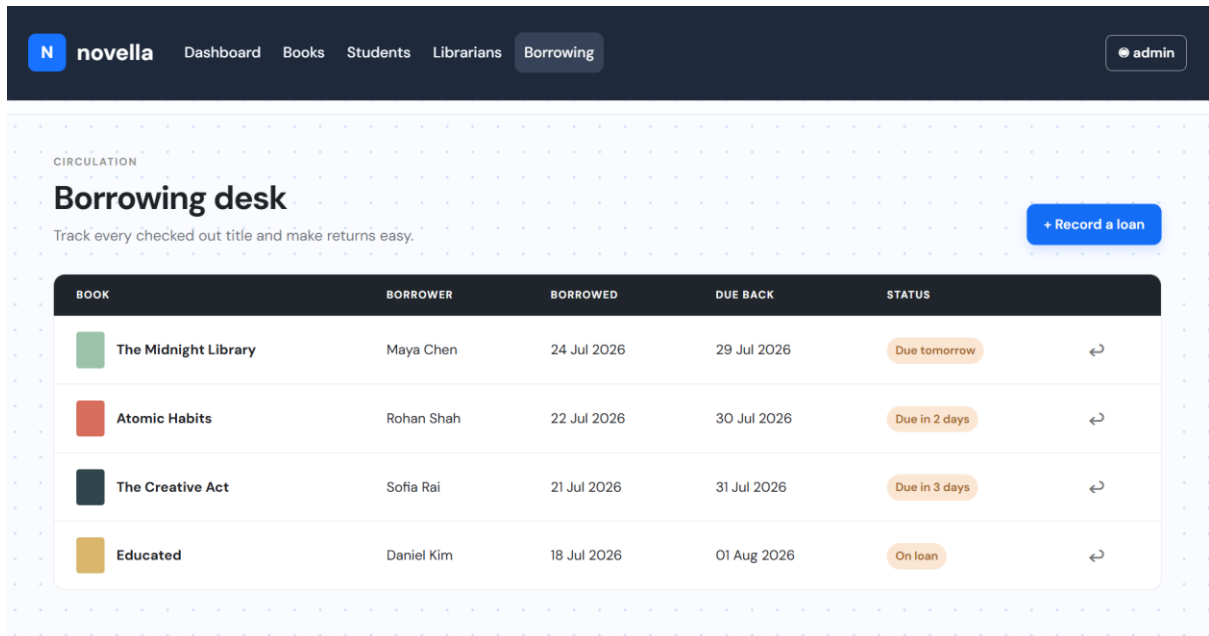


Figure 11.9 – Return Book

Allows the administrator to mark borrowed books as returned.

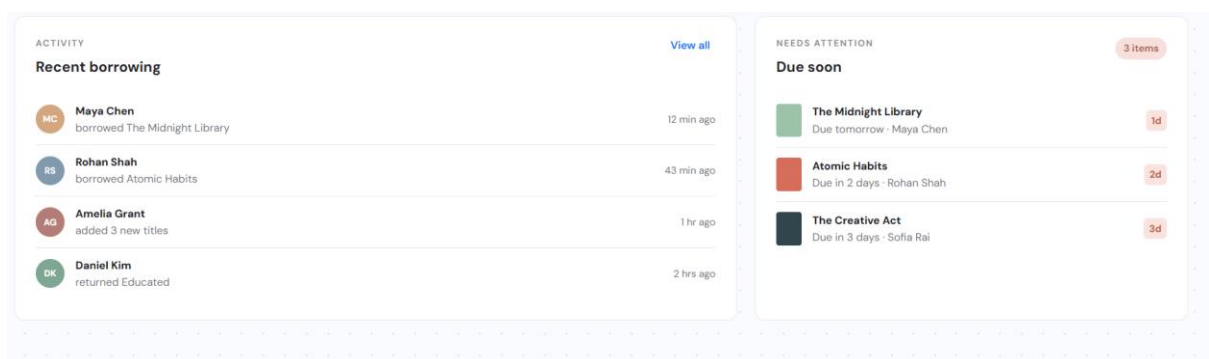
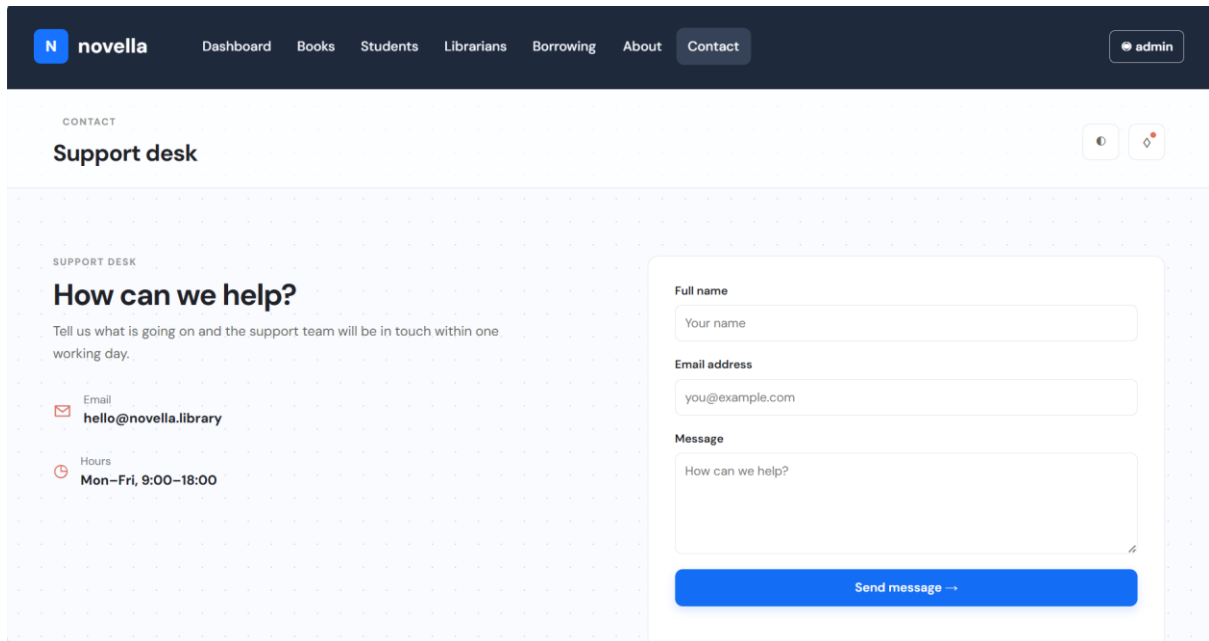


Figure 11.10 – Contact Page

Displays contact information and support details.



The screenshot shows a web application interface for a library named 'novella'. The top navigation bar is dark blue with the 'novella' logo on the left and a list of menu items: Dashboard, Books, Students, Librarians, Borrowing, About, and Contact. The 'Contact' item is highlighted. On the right side of the navigation bar, there is a user profile icon labeled 'admin'. Below the navigation bar, the page has a light blue background with a subtle grid pattern. The main heading is 'Support desk'. To the left of the main content area, there is a sidebar with contact information: 'Email hello@novella.library' and 'Hours Mon–Fri, 9:00–18:00'. The main content area features a large heading 'How can we help?' followed by a subheading 'Tell us what is going on and the support team will be in touch within one working day.' To the right of this text is a form with three input fields: 'Full name' (placeholder: 'Your name'), 'Email address' (placeholder: 'you@example.com'), and 'Message' (placeholder: 'How can we help?'). Below the form is a blue button labeled 'Send message →'.

CHAPTER -12

FUTURE ENHANCEMENTS

While the current implementation fulfills all primary objectives, several enhancements can be incorporated in future iterations to further improve the system:

12.1 Database Integration

- Migrate from in-memory arrays to MongoDB (NoSQL) or PostgreSQL (SQL) for persistent data storage.
- Implement Mongoose ODM or Sequelize ORM for structured data access patterns.
- Add database indexing for optimized search queries.

12.2 Advanced Features

- Real-time notifications using WebSockets for borrow/return events.
- Email notifications for overdue books using Nodemailer.
- Barcode/QR code scanning for rapid book identification.
- PDF report generation for administrative analytics.
- Multi-language support (i18n) for Hindi, English, and other regional languages.

12.3 Security Enhancements

- Implement bcrypt for stronger password hashing.
- Add CSRF (Cross-Site Request Forgery) token protection.
- Implement rate limiting to prevent brute-force attacks.
- Add JWT (JSON Web Token) based API authentication for mobile clients.

12.4 Deployment

- Deploy to cloud platforms such as Vercel, Railway, or AWS EC2.
- Implement CI/CD pipelines using GitHub Actions.
- Add Docker containerization for consistent deployment environments.

CHAPTER – 13

CONCLUSION

The Library Management System (LMSystem) project has been successfully designed, developed, and deployed as a comprehensive web application that fulfills all the stated objectives of the MPonline Advanced Software Engineering & Development Internship program.

Throughout the development process, several key competencies were demonstrated and strengthened:

- **Full-Stack Web Development:** The project demonstrates end-to-end development proficiency, using ASP.NET Core MVC, C#, Entity Framework Core, SQL Server, HTML, CSS, and Bootstrap.
- **Software Architecture:** The MVC pattern implementation ensures clean separation of concerns, making the codebase maintainable and extensible for future enhancements.
- **UI/UX Design Excellence:** The custom CSS design system with dual-theme support showcases advanced frontend engineering capabilities, including CSS custom properties, responsive design, and premium visual aesthetics.
- **Security Implementation:** Session-based authentication, password hashing, and route protection middleware demonstrate awareness of web application security best practices.
- **Data Management:** Complete CRUD operations across six entity modules with search, filter, and export capabilities highlight data manipulation proficiency.
- **Testing** –The testing process confirmed that all modules perform according to the specified requirements. The application is reliable, user-friendly, and suitable for deployment in schools, colleges, and small libraries.

This internship project has been an invaluable learning experience, providing hands-on exposure to modern web development practices, version control with Git and GitHub, and the iterative development process of building a polished, feature-complete application.

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